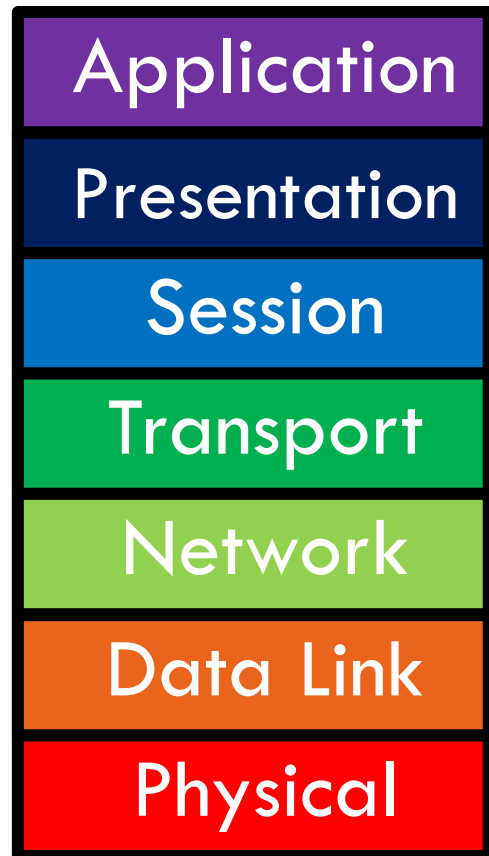


Physical Layer



Physical Layer



□ Function:

- Get bits across a physical medium

Physical Layer

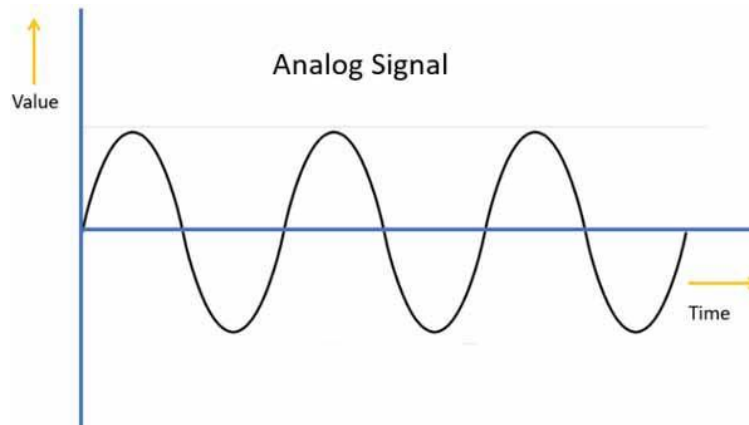
- The physical layer is the **first and lowest layer** of the Open System Interconnection Model (OSI Model.)
- The physical layer (also known as layer 1) **deals with bit-level transmission** between different devices and **supports electrical or mechanical interfaces** connecting to the physical medium for **synchronized communication**.
- This layer plays with most of the network's physical connections—wireless transmission, cabling, connectors and types, network interface cards, and more—as per network requirements.

Introduction

1. Analog & Digital Signals
2. Bit Rate & Bit Length

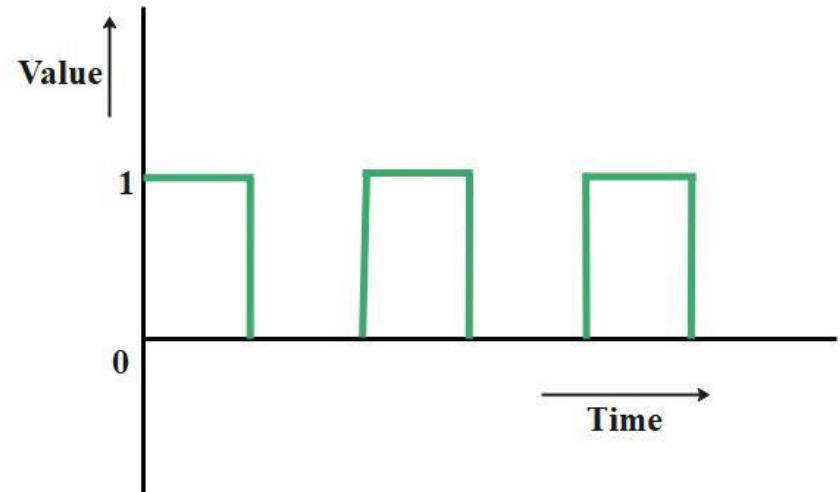
Analog Signals

- An analog signal is a **continuous signal** that changes smoothly over time.
- Characteristics:
 - ▣ Continuous in both time and amplitude
 - ▣ Can **have infinite values**
 - ▣ **More affected by noise**
 - ▣ Used in traditional telephone systems, radio, and TV broadcasting



Digital Signals

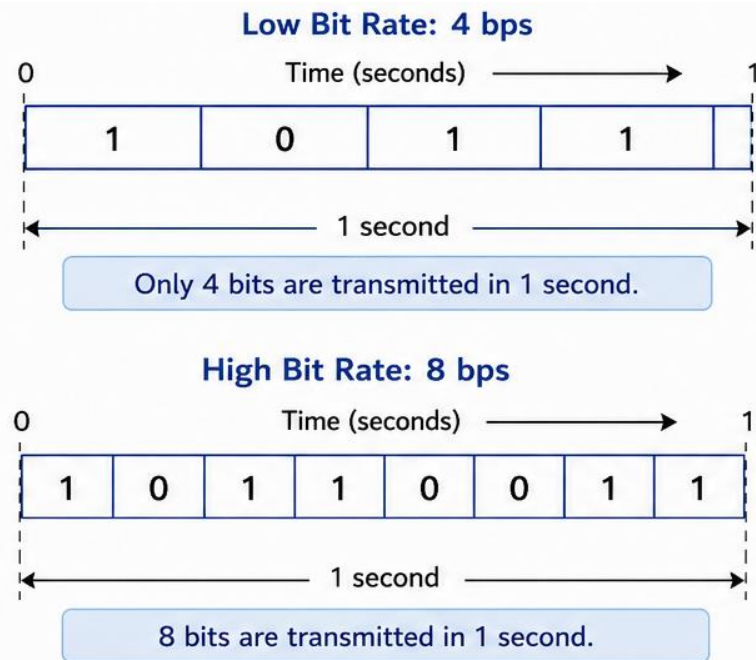
- A digital signal **has only discrete values**, usually:
 - ▣ 0 (LOW)
 - ▣ 1 (HIGH)
- Characteristics
 - ▣ Discrete in amplitude
 - ▣ **Less affected by noise**
 - ▣ Easier to store and process
 - ▣ Used in computers and the Internet



Bit Rate

- Bit Rate is the **number of bits transmitted per second** (bps).
- Example:

An Internet connection of 100 Mbps means it can transmit approximately 100 million bits every second.

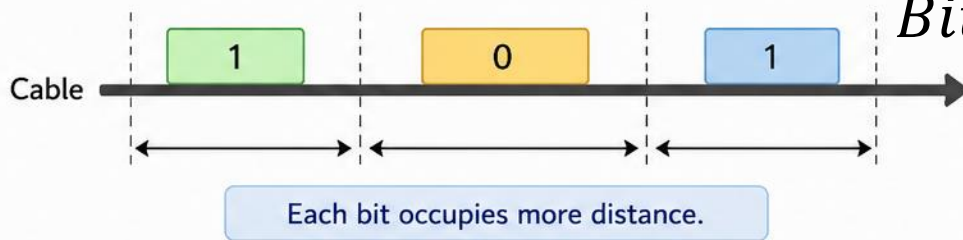


$$\text{Bit Rate} = \frac{\text{Number of bits}}{\text{Time}}$$

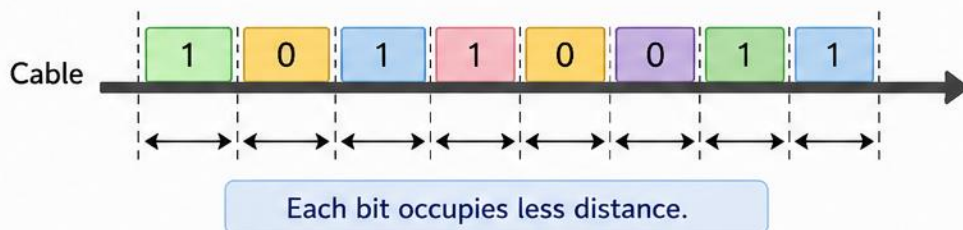
Bit Length

- Bit Length is the **physical distance occupied by one bit on the transmission medium** (cable).
- It tells us how much cable length corresponds to one bit.

Low Bit Rate → Large Bit Length



High Bit Rate → Small Bit Length



$$\text{Bit Length} = \frac{\text{Propagation Speed}}{\text{Bit Rate}}$$

Why both are needed ?

Higher Bit Rate $\Rightarrow$ More bits each second $\Rightarrow$ Bits are packed closer together $\Rightarrow$ Smaller Bit Length

- ❑ Network Engineer installed 100-meter Ethernet cable between two office buildings that carries signal at 100Mbps.
- ❑ Say company wants to upgrade to 1 Gbps.
- ❑ $\therefore$ Bit length calculated for both.
 - ▣ 100Mbps has each bit 2 meters apart.
 - ▣ 1 Gbps has each bit 20 centimeters apart.
- ❑ Now, the engineer has to check for the cable quality to know if it can handle the new metrics of communication to avoid any signal distortion/inter-symbol interference.
- ❑ Doing so will avoid slow file transfers/frequent retransmissions/network disconnects etc.



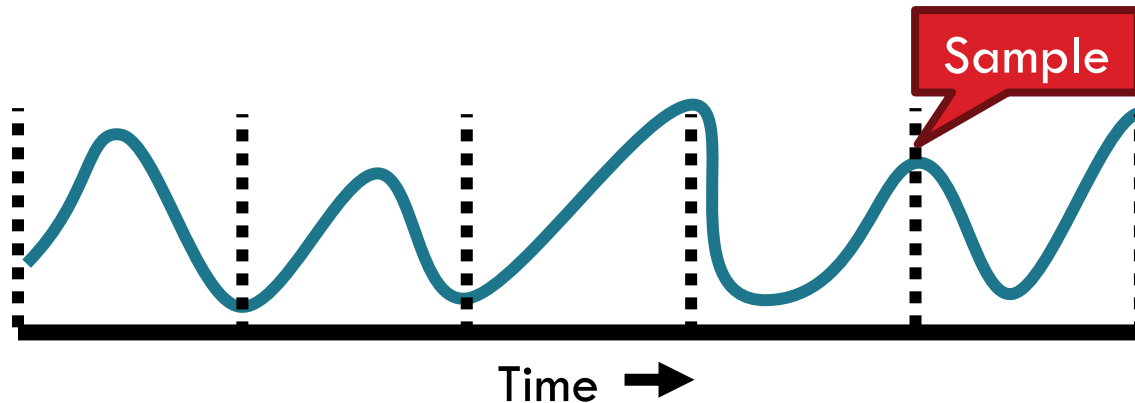
Data Encoding Techniques:

Digital Modulation ;
process of
converting bits to
signal & vice versa



Assumptions

- We have two discrete signals, high and low, to encode 1 and 0
- Transmission is **synchronous**, i.e., there is a clock that controls signal sampling

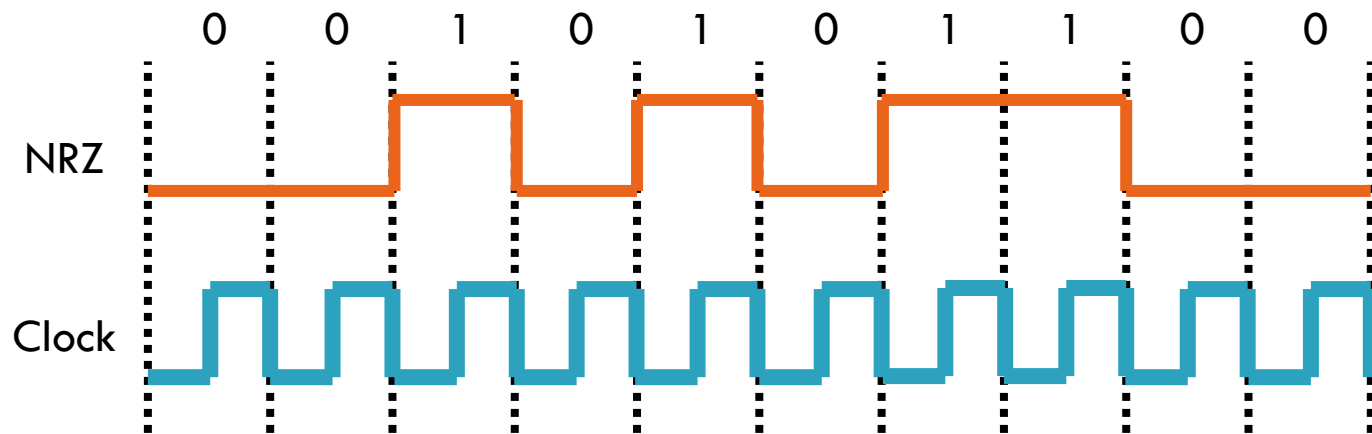


- Amplitude and duration of signal must be significant



Non-Return to Zero (NRZ)

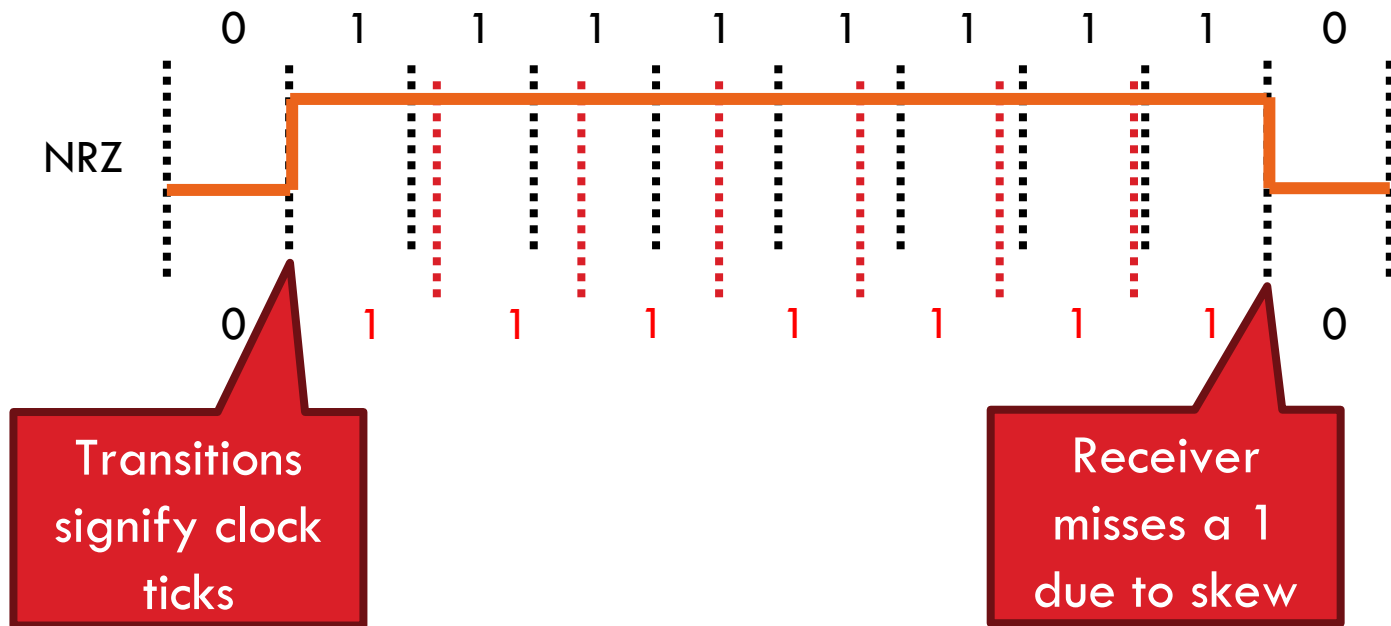
- 1 → high signal, 0 → low signal



- Problem: long strings of 0 or 1 cause desynchronization
 - ▣ How to distinguish lots of 0s from no signal?
 - ▣ How to recover the clock during lots of 1s?

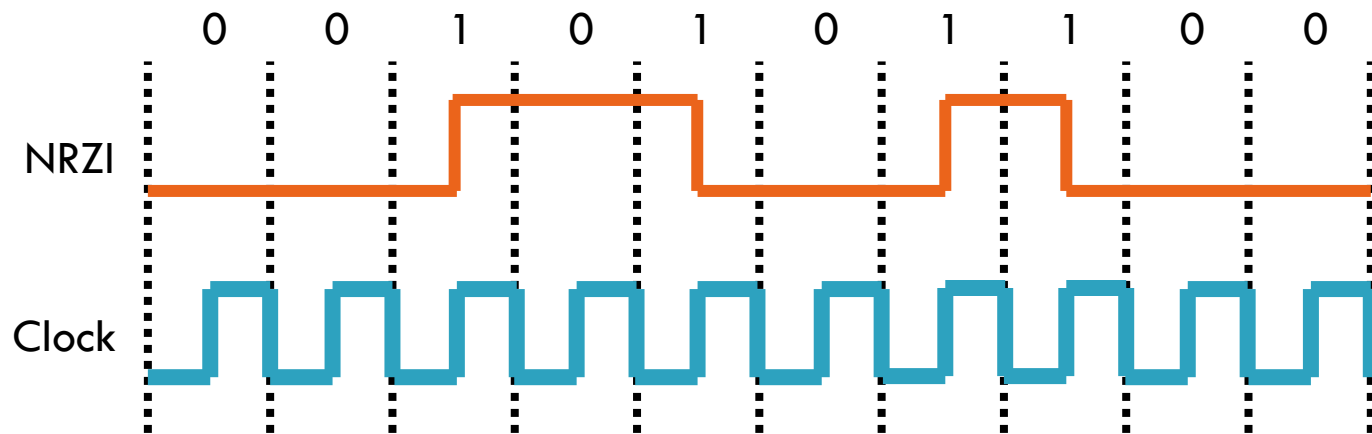
Desynchronization

- Problem: how to recover the clock during sequences of 0's or 1's?



Non-Return to Zero Inverted (NRZI)

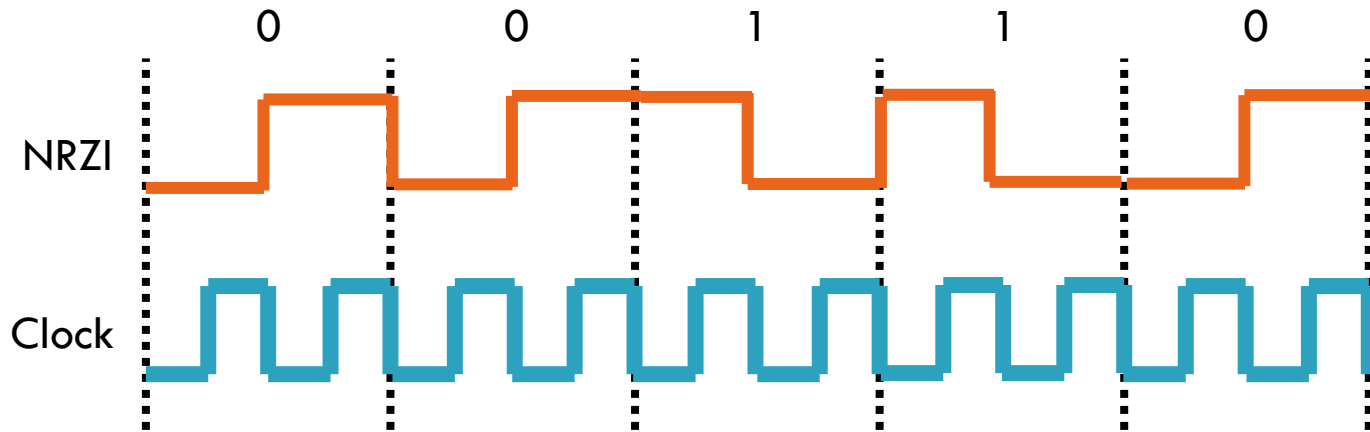
- 1 → make transition, 0 → remain the same



- Solves the problem for sequences of 1s, but not 0s

Manchester

- 1 → high-to-low, 0 → low-to-high



- Good: Solves clock skew (every bit is a transition)
- Bad: Halves throughput (two clock cycles per bit)

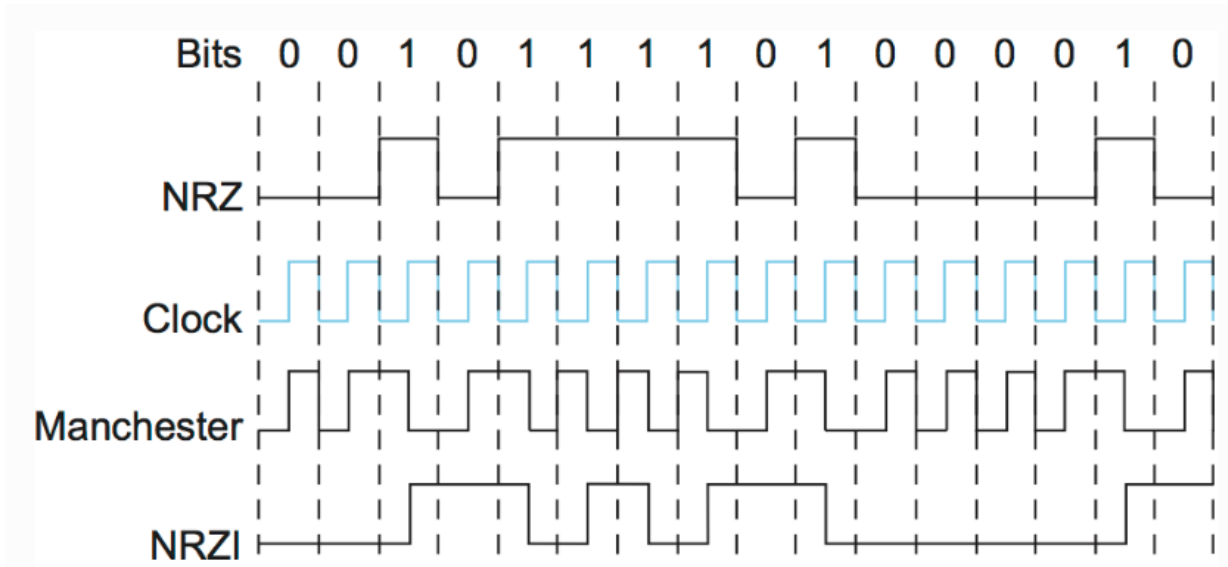
4-bit/5-bit (100 Mbps Ethernet)

8-bit / 10-bit used in Gigabit Ethernet

- Observation: NRZI works as long as no sequences of 0
- Idea: encode all 4-bit sequences as 5-bit sequences **with no more than one leading 0 and two trailing 0**

4-bit	5-bit	4-bit	5-bit
0000	11110	1000	10010
0001	01001	1001	10011
0010	10100	1010	10110
0011	10101	1011	10111
0100	01010	1100	11010
0101	01011	1101	11011
0110	01110	1110	11100
0111	01111	1111	11101

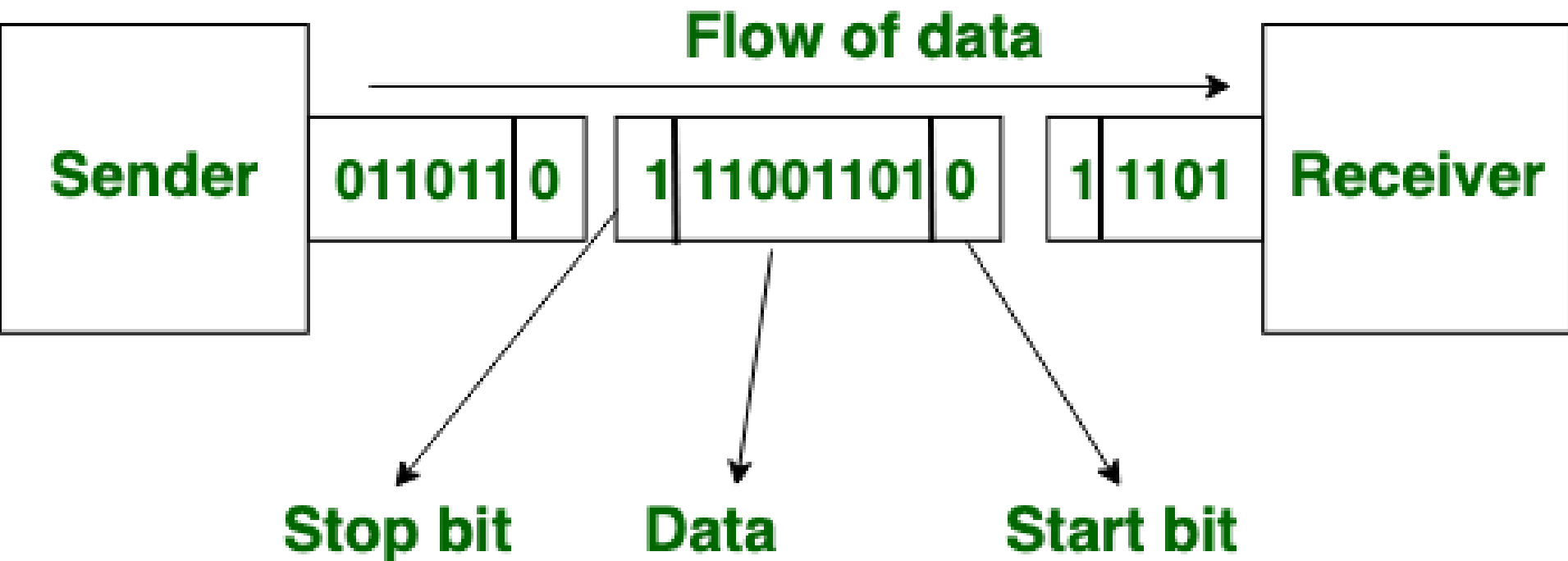
Diff Encoding Strategies



Asynchronous & Synchronous Transmission

Asynchronous Transmission

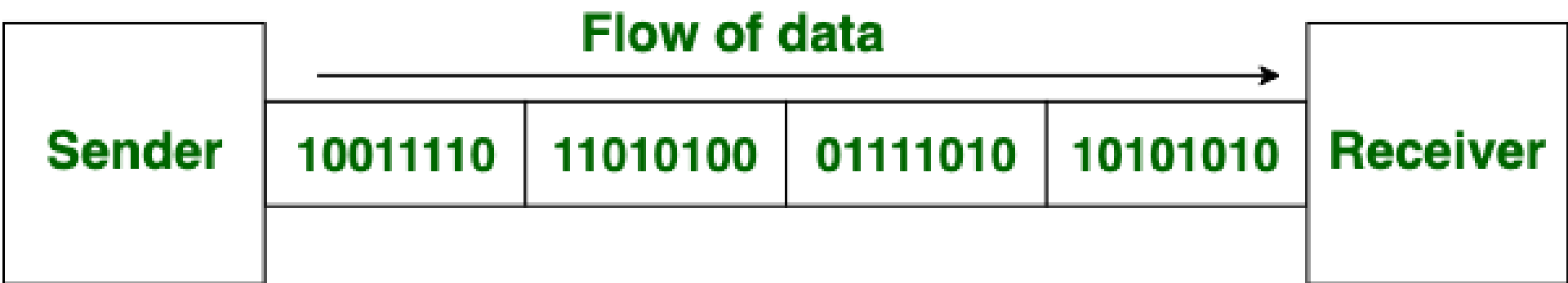
- ❑ Data is **sent in form of byte** or character.
- ❑ This transmission is the **half-duplex type transmission**.
- ❑ In this transmission **start bits and stop bits are added** with data.
- ❑ It **does not require synchronization**.
 - ▣ The sender and receiver do not share a common clock signal.
- ❑ Asynchronous transmission is like sending individual text messages without knowing exactly when the other person will read them.
- ❑ Example:
 - ▣ Email
 - ▣ Forums



Asynchronous Transmission

Synchronous Transmission

- ❑ Data is **sent in the form of blocks or frames**.
- ❑ This transmission is the **full-duplex type**.
- ❑ Data is **sent in a continuous stream**, with each byte or chunk of data **following the previous one without any gaps**.
- ❑ Between sender and receiver, **synchronization is compulsory**.
 - ▣ Both the sender and receiver are synchronized with a common clock signal.
- ❑ Example:
 - ▣ Chat Rooms
 - ▣ Telephonic Conversations
 - ▣ Video Conferencing



Synchronous Transmission

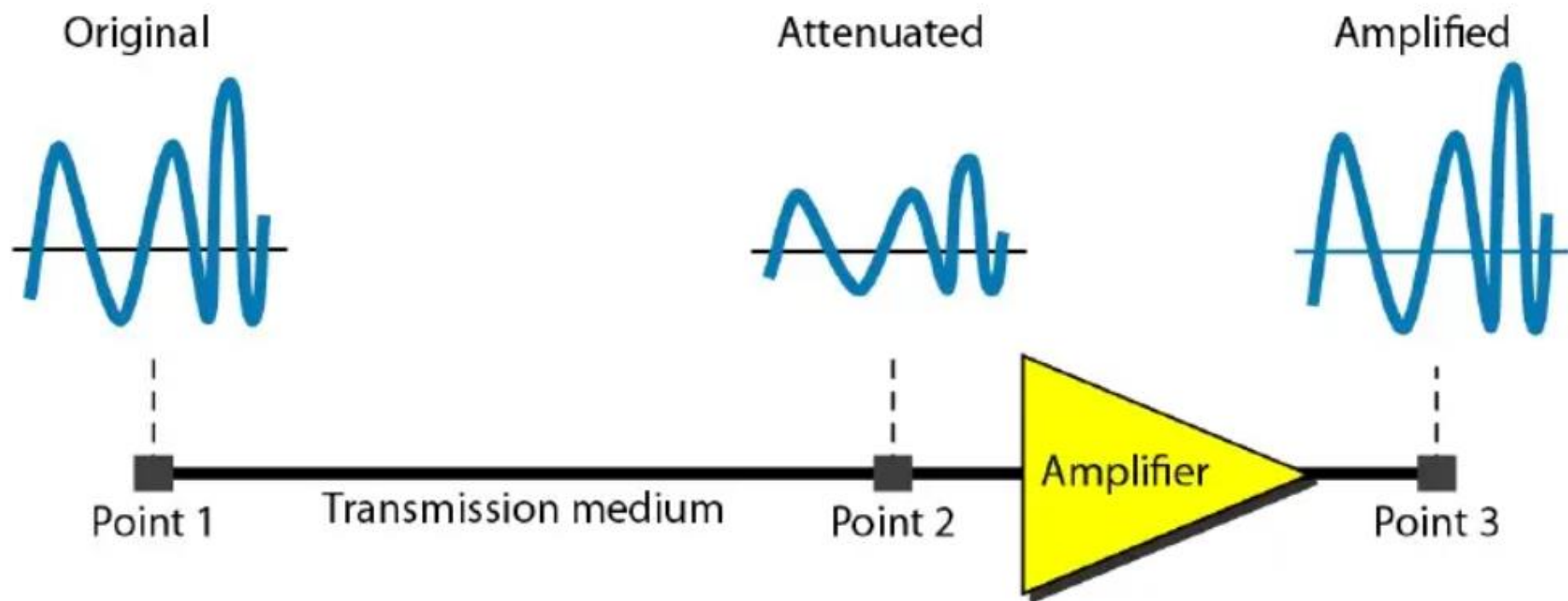
Transmission Impairments

1. Attenuation
2. Distortion
3. Noise

Attenuation



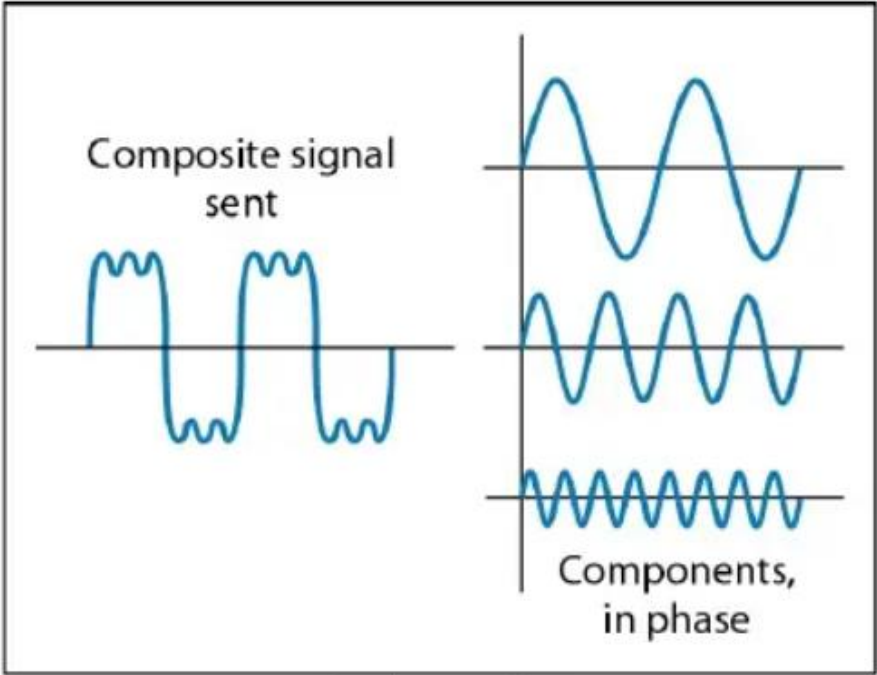
- Means **loss of energy** > **weaker signal**
- When a signal travels through a medium it loses energy over coming the resistance of the medium
- **Amplifiers are used** to compensate for this loss of energy by amplifying the signal.



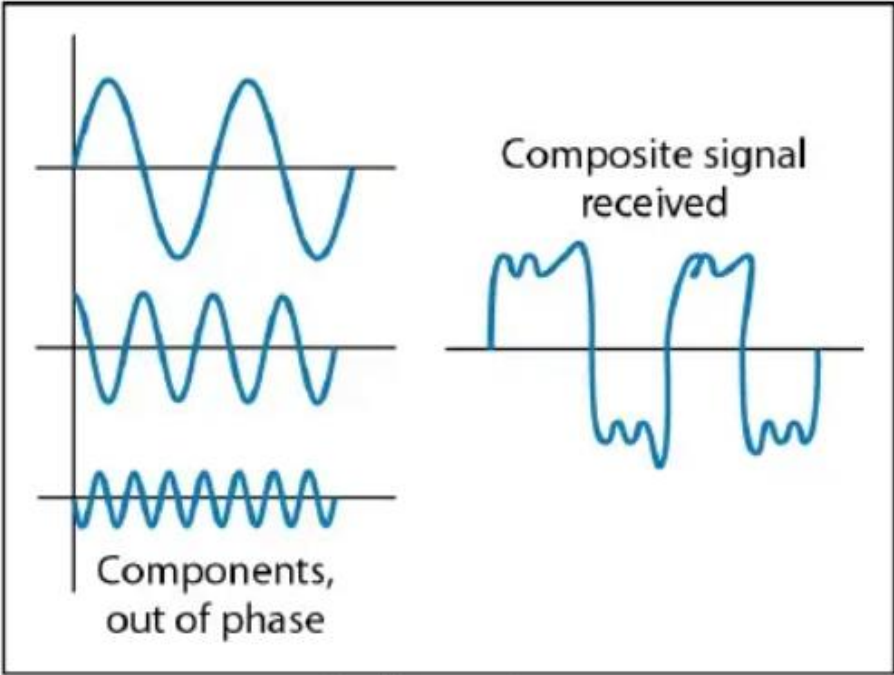
Distortion



- Means that the **signal changes its form or shape**
- Distortion **occurs in composite signals**
- Each frequency component has its own propagation speed traveling through a medium.
- The different components therefore arrive with different delays at the receiver.



At the sender



At the receiver

Noise

There are different types of noise

- **Thermal –**

- ▣ Random noise of electrons in the wire creates an extra signal

- **Induced –**

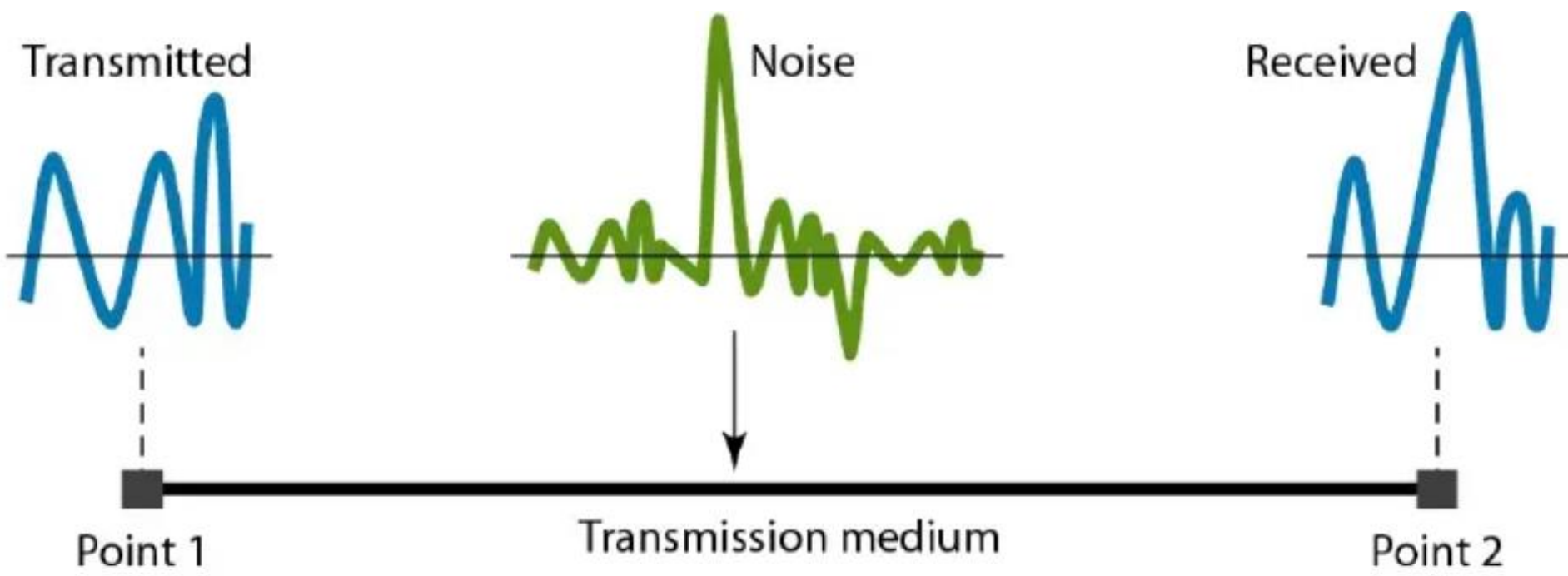
- ▣ from motors and appliances, devices act as transmitter antenna and medium as receiving antenna.

- **Crosstalk –**

- ▣ Same as above but between two wires.

- **Impulse –**

- ▣ Spikes that result from power lines, lightning, etc.



QUIZ



A weather station continuously records the outside temperature throughout the day. Which type of signal best represents the changing temperature?

A. Digital signal

B. Analog signal

C. Binary signal

D. Hexadecimal signal



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A shopping mall upgrades from analog CCTV cameras to IP (digital) cameras. What is one major advantage?

- A. Digital cameras cannot be connected to a network.
- B. Digital cameras support remote monitoring over the Internet.
- C. Analog cameras provide higher-quality images.
- D. Analog cameras require less maintenance because they don't use electricity.



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Ananya downloads a movie using a 100 Mbps internet connection. What does 100 Mbps represent?

- A. Number of bytes transferred every second
- B. Number of bits transferred every second
- C. Size of the movie
- D. Distance travelled by data



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A telecom company lays a 50 km optical fiber cable between two cities. Which statement about bit length is correct?

- A. Bit length depends on the physical distance occupied by one bit in the transmission medium.
- B. Bit length is the number of bits sent every second.
- C. Bit length is always 1 meter.
- D. Bit length is the size of a computer file.



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A student says: "Since Wi-Fi uses radio waves, the Internet transmits analog data." Which statement is correct?

- A. Correct, because radio waves are analog.
- B. Incorrect, because the Internet transmits digital data, while radio waves act only as the carrier.
- C. Correct, because all wireless communication is analog.
- D. Incorrect, because Wi-Fi uses light signals.



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Which of the following is naturally an analog signal but is usually transmitted digitally over today's Internet?

- A. PDF file
- B. Human voice during a WhatsApp call
- C. Text message
- D. Excel spreadsheet



Which of the following is naturally an analog signal but is usually transmitted digitally over today's Internet?

A. PDF file

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Which system still commonly transmits analog radio signals carrying information?

- A. FM Radio
- B. Ethernet
- C. Optical Fiber
- D. USB Cable



Which system still commonly transmits analog radio signals carrying information?

A. FM Radio

B. Ethernet

C. Optical Fiber

D. USB Cable



Which statement is correct?

- A. Hz and bps measure the same quantity.
- B. Hz measures frequency; bps measures data rate.
- C. Hz is always greater than bps.
- D. Both are units of bandwidth.



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Nyquist Formula

- Formula:

$$\textit{Maximum Bit Rate} = 2B \log_2 L$$

Where, B : Bandwidth (in Hertz)

L : Number of signal levels

- Max. number of bits per second (**bit rate**) over a **noiseless channel**.
- **High-quality optical fiber** installed in a laboratory where there is **almost no interference**.

Shannon Formula

□ Formula:

$$\text{Channel capacity} = B \log_2(1 + SNR)$$

Where, B = Bandwidth (in Hertz)

SNR = Signal-to-Noise ratio (not in dB, to be converted from dB)

$$SNR_{linear} = 10^{\frac{SNR(in\ dB)}{10}}$$

- Maximum data rate (channel capacity) that can be achieved without errors over a noisy channel.
- Using Wi-Fi in a crowded mall. Many routers, Bluetooth devices, and phones create interference.

NYQUIST vs SHANNON FORMULA

NYQUIST FORMULA (Noiseless Channel)



Applies to an ideal channel where **NO NOISE** is present.

$$\text{Maximum Bit Rate } (R_{\max}) = 2B \log_2 L$$

Where,

B = Bandwidth of the channel (in Hz)

L = Number of signal levels

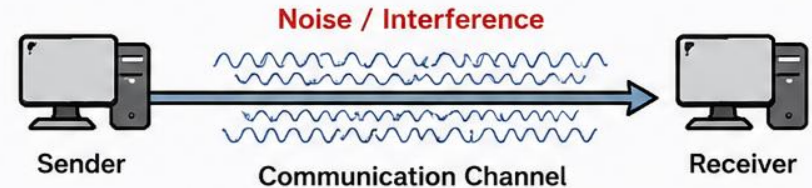
Key Points



- Depends on Bandwidth (B) and Number of Signal Levels (L)
- Gives the theoretical maximum bit rate for noiseless channel

VS

SHANNON FORMULA (Noisy Channel)



Applies to a real channel where **NOISE** is present.

$$\text{Channel Capacity } (C) = B \log_2 (1 + \text{SNR})$$

Where,

B = Bandwidth of the channel (in Hz)

SNR = Signal to Noise Ratio (S / N) [in linear scale]

Key Points



- Depends on Bandwidth (B) and Signal to Noise Ratio (SNR)
- Gives the maximum channel capacity in the presence of noise



In short : **Nyquist** → No Noise | **Shannon** → Noise is present

Transmission Medium

It actually lies below the physical layer & is controlled directly by the physical layer.

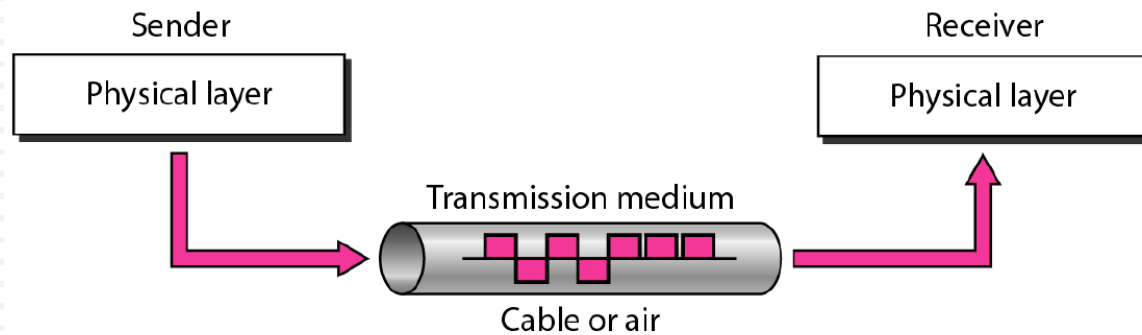


Figure *Transmission medium and physical layer*

Classes of Transmission Medium

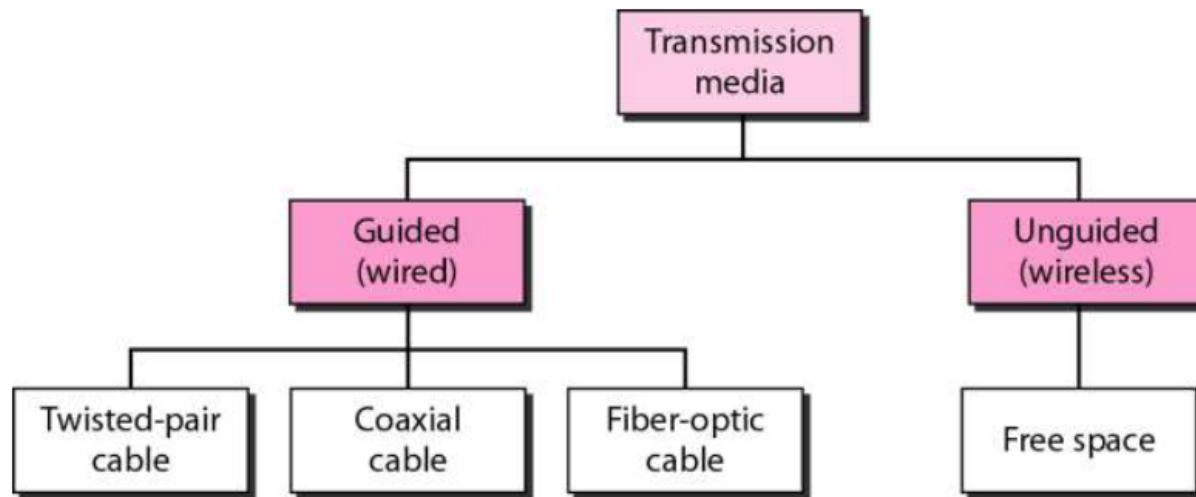


Figure : *Classes of transmission media*

Guided Transmission Medium

1. Twisted Pair Cable

- It has **2 conductors** (mostly copper) **insulated individually**.
- **One** carries signals to the receiver & the **other** is only used as a ground reference.
- Two kinds : Unshielded Twisted Pair cable (**UTP cable**); Shielded Twisted Pair cable (**STP cable**).
- **Metal casing in STP cable**, improves the quality of cable (**by preventing the penetration of noise/crosstalk**) but is bulkier and expensive.
- **Uses RJ45 connector** [RJ = Registered Jack]

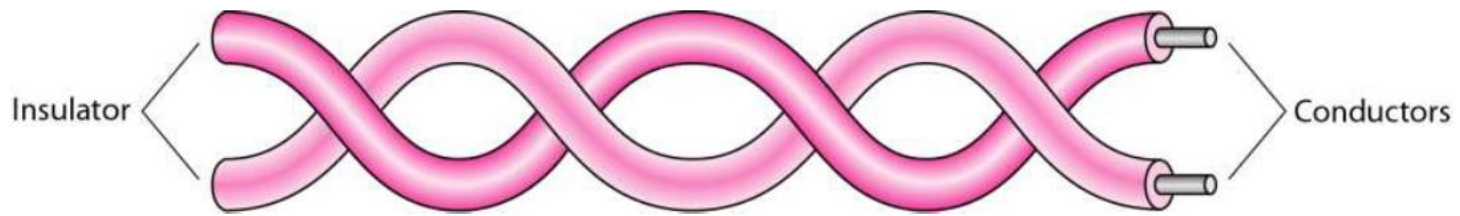
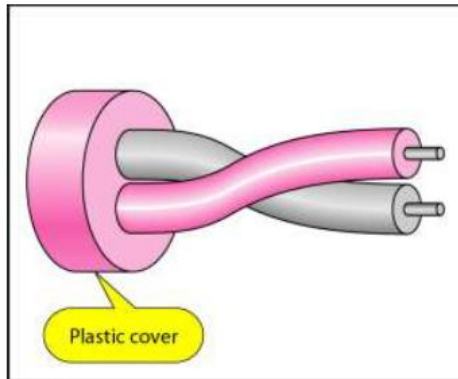
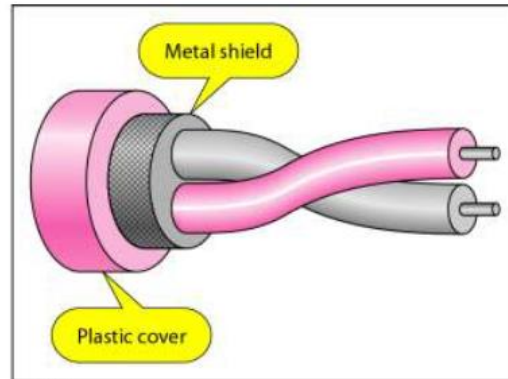


Figure: Twisted-pair cable

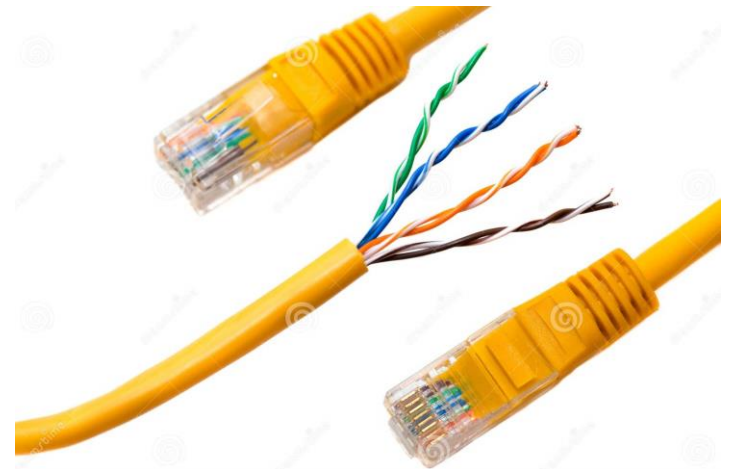


a. UTP



b. STP

Figure: UTP and STP cables



Twisted cable with RJ45 connector

Applications.

- ❑ Telephone lines.
- ❑ LAN with low cost, low performance applications.

Types of Twisted Pair Cable

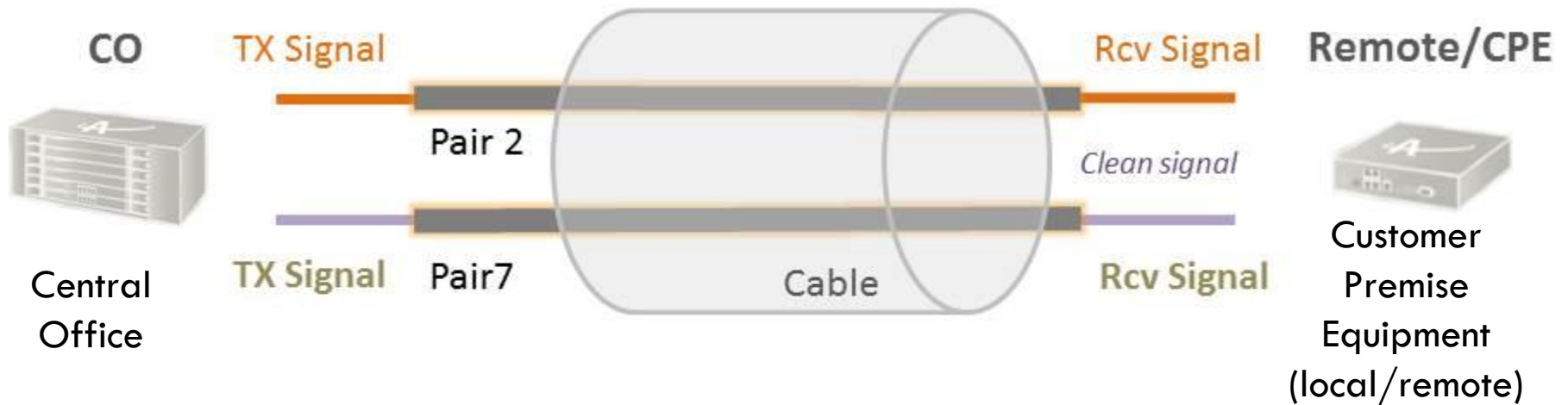
Unshielded Twisted Pair (UTP):

- ❑ UTP does not have the ability to block electromagnetic interference.
- ❑ The outer insulating jacket **protects** the cable **from physical stress or damage**.
- ❑ Application: most common form of network cabling for workgroups and **departmental local area networks (LANs)**.
- ❑ Advantage: Least expensive, Easy to install
- ❑ Disadvantage: **Short distance transmission** due to attenuation

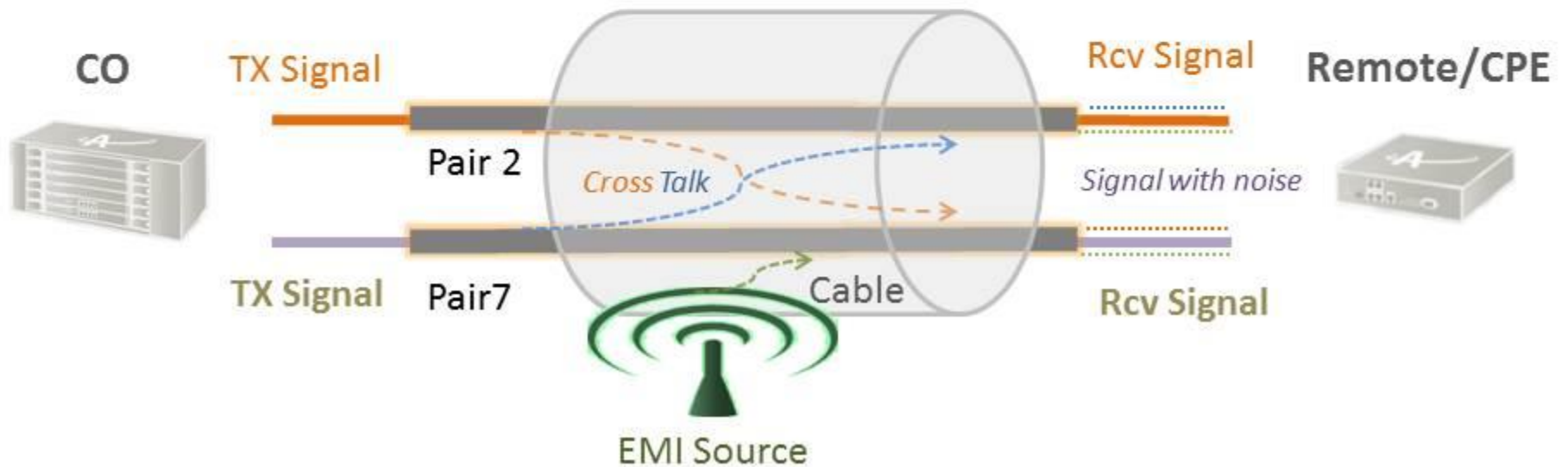
Shielded Twisted Pair (STP):

- ❑ STP consists of a special jacket (a copper braid covering or a foil shield) **to block electromagnetic interference & crosstalk**.
- ❑ Application: **fast-data-rate** Ethernet, voice and data channels of telephone lines.
- ❑ Advantage: **Eliminates crosstalk**, comparatively faster
- ❑ Disadvantage: More expensive, bulky, difficult to install.

Transmission In "Ideal" Environment Without Noise



Impact of EMI and Crosstalk (without mitigation and cancellation)



2. Coaxial Cable

- ❑ Carries **signals of higher frequency** ranges **than in TP** cables.
- ❑ Has conductor in an insulating sheath, encased in a conductor of metal foil/braid/combination of both, again encased in an insulating sheath, thereby protected by plastic cover.

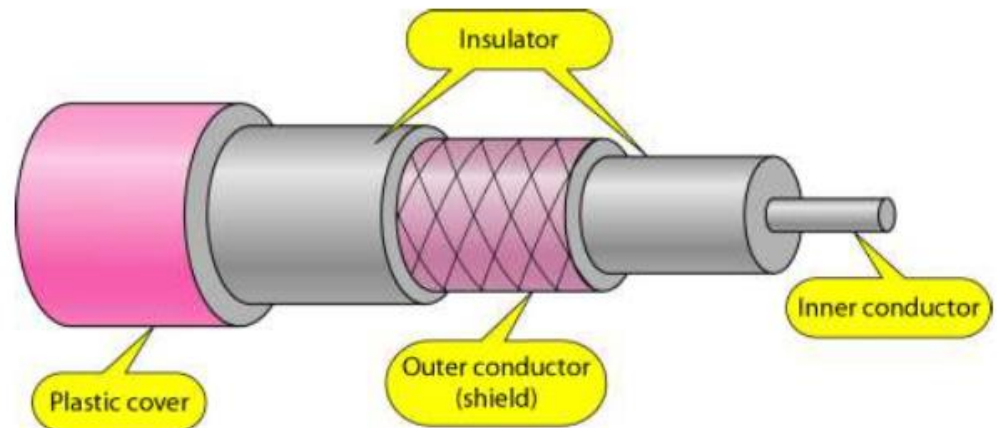


Figure Coaxial cable

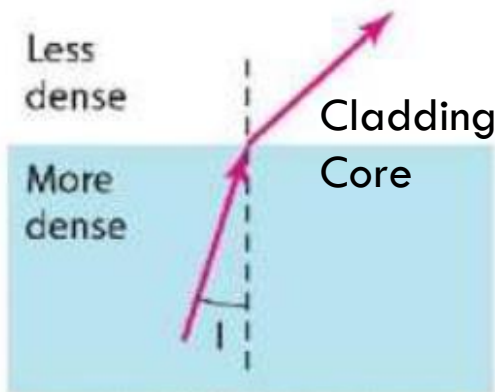
Advantages & Applications



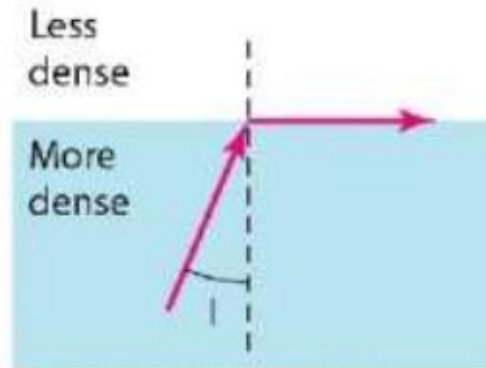
- ❑ Has **higher bandwidth**.
- ❑ Has excellent **noise immunity**.
- ❑ Can be used for **longer distances** at higher data rates.
- ❑ Can be used **for both analog and digital data transmission**.
- ❑ **Inexpensive compared to fibre optic cables**.
- ❑ Used in LAN + Television distribution.

3. Fibre Optic Cable

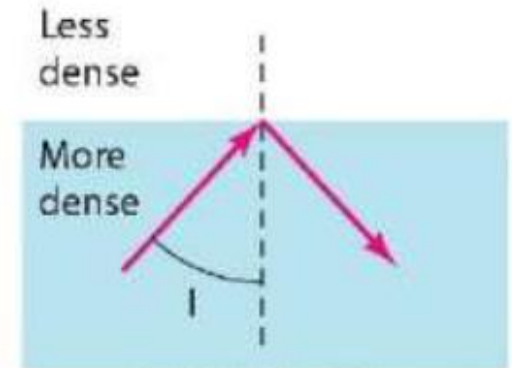
- It is made of glass/plastic & transmits light signals.
- If ray of light travels through one substance to the next substance, it changes its direction as it passes the interface between the two substances.
- **Incidence angle (I):** Angle of incidence of light on the core-cladding interface
- **Critical angle:** Angle of incidence of light on the core-cladding interface beyond which the light undergoes total internal reflection.



$I < \text{critical angle}$,
refraction



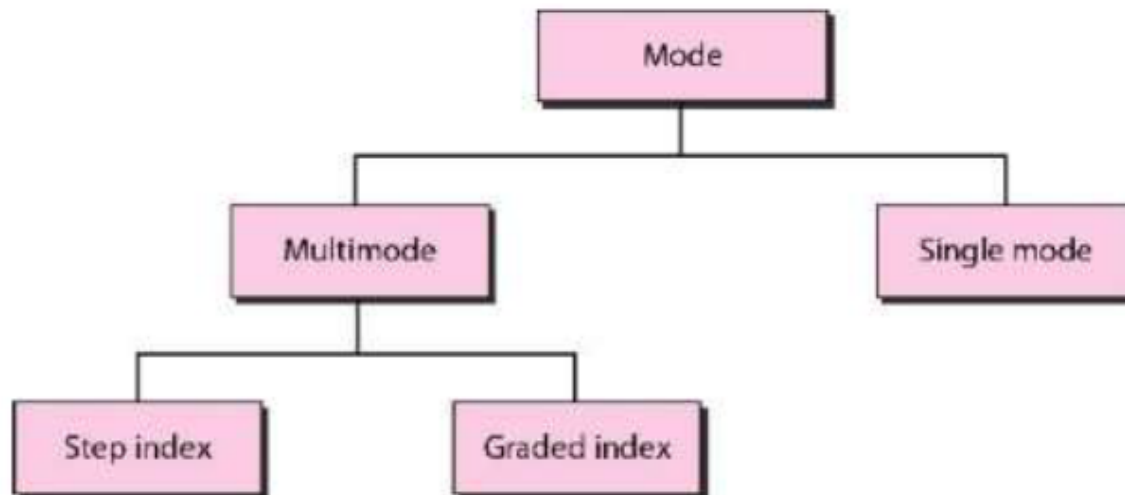
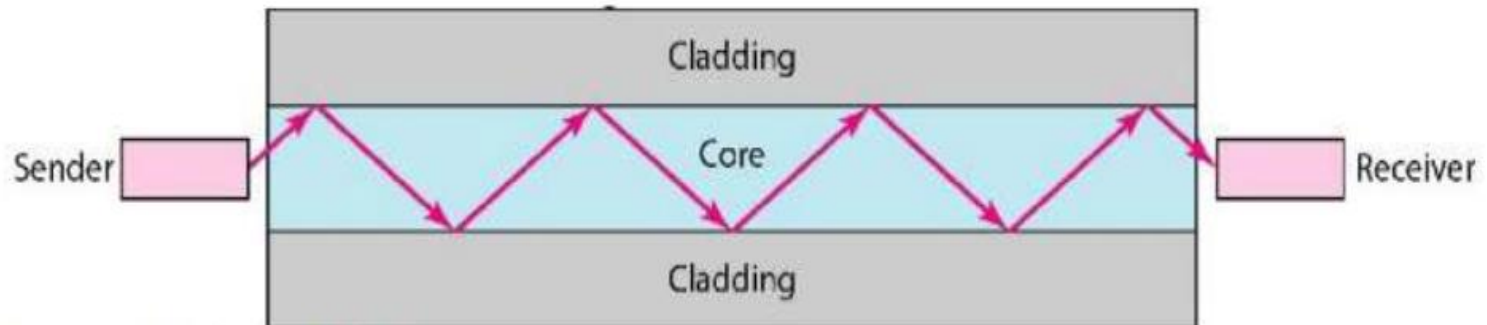
$I = \text{critical angle}$,
refraction



$I > \text{critical angle}$,
reflection

Propagation Modes

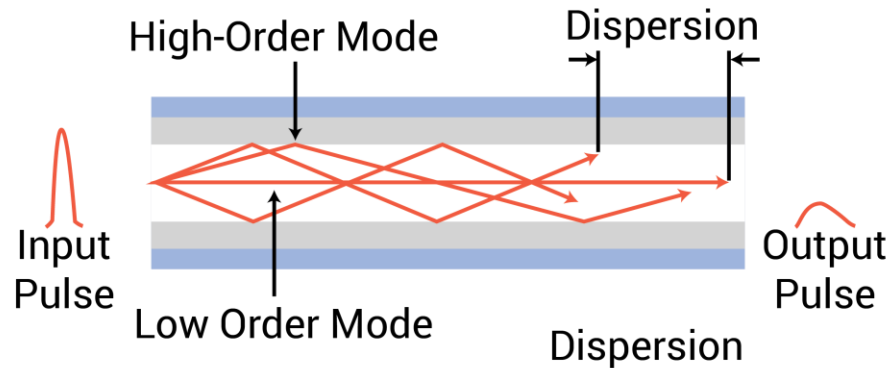
Optical fibres = glass/plastic core surrounded by cladding of less dense glass/plastic.



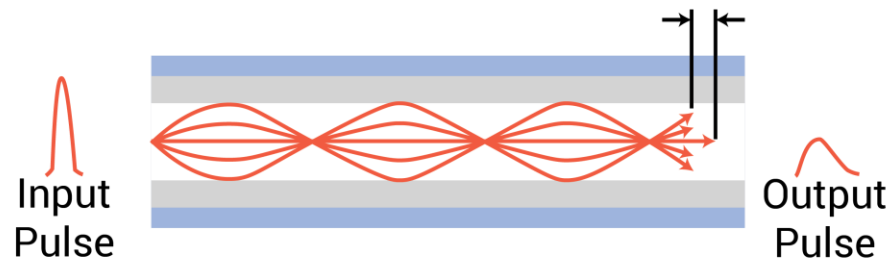
Propagation Modes

Optic Fiber Signal Index Types

Multimode Step Index



Multimode Grade Index



Singlemode Mono



Advantages, Disadvantages, Applications

Advantages

- ❑ Higher bandwidth.
- ❑ Less signal attenuation.
- ❑ Light weight.
- ❑ Immunity to electromagnetic interference.

Disadvantages

- ❑ Installation & maintenance is high.
- ❑ Unidirectional light propagation.
- ❑ Cables/interfaces more expensive.

Applications

- ❑ Mainly for backbone networks.
- ❑ Currently, also for home networks: fiber-to-the-home (FTTH) networks (due to demand for Higher Bandwidth for video streaming, online gaming, and cloud services etc., cost has decreased significantly over the years, making it more affordable for home users, high-speed internet access etc.)

Unguided Transmission Medium

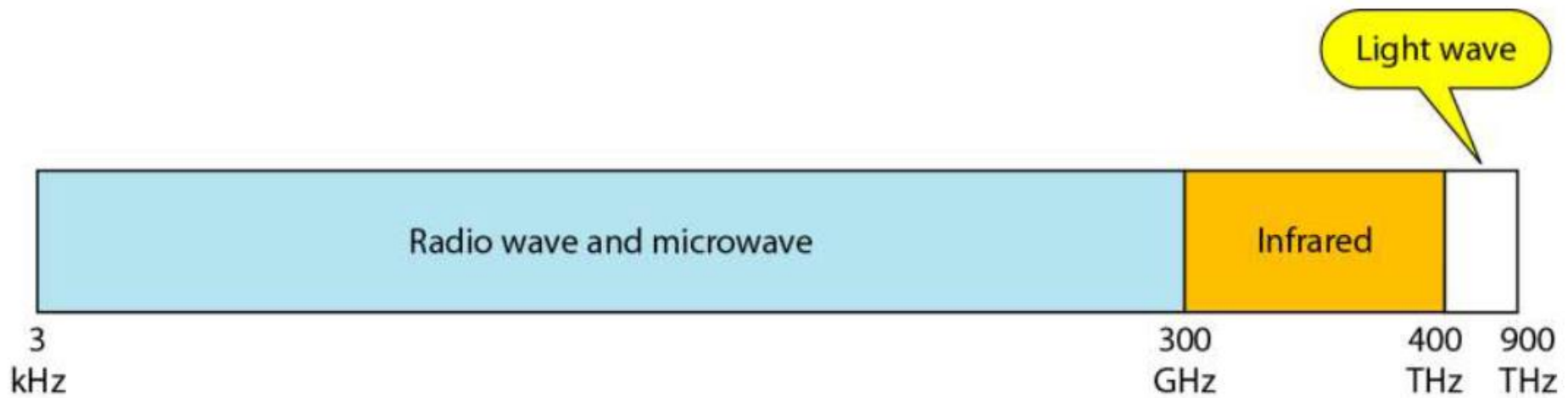


Figure: *Electromagnetic spectrum for wireless communication*

Propagation Methods

- **Ground Propagation** : **radio waves** (low frequency) travel along the curvature of earth.
- **Sky propagation** : **radio waves** (high frequency) radiate to ionosphere > get reflected back to earth.
- **Line of sight (LoS) propagation** : high frequency signals in straight line from antenna to antenna.

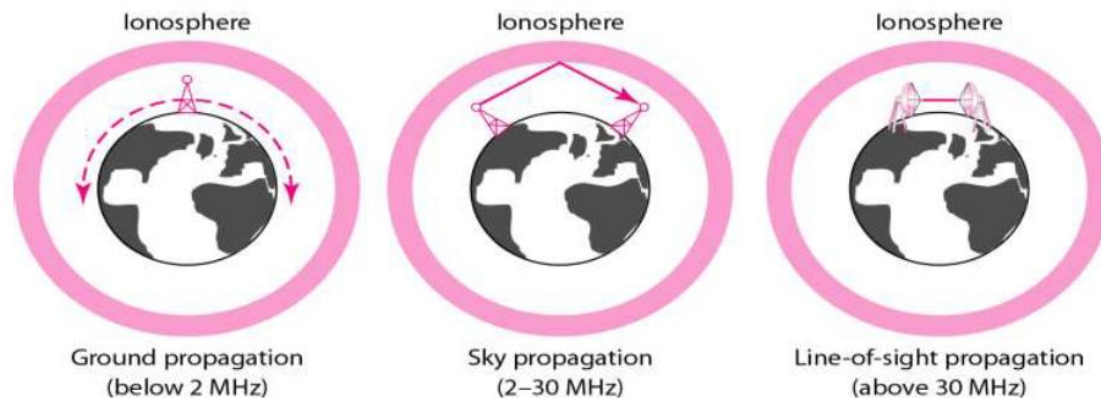


Figure Propagation methods

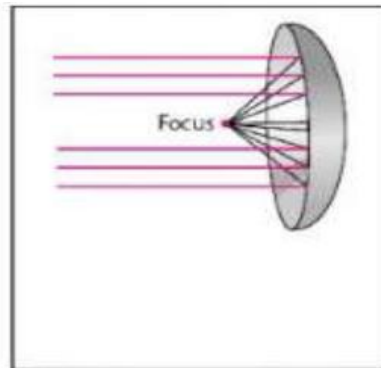
1. Radio Waves

- Electromagnetic waves ranging in frequencies between 3kHz – 1 GHz.
- Mostly **omni-directional**. [propagated in all directions]
- Since they can travel long distances, they are a good candidate for **long-distance broadcasting i.e. AM radio** (used in weather warnings, Amber Alerts etc.).
- Can **penetrate through walls**.
- Applications : multicast communications [One sender, many receivers] like AM & FM radio, paging systems etc.

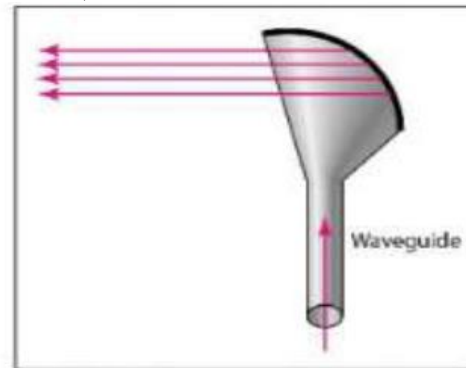


2. Microwaves

- Electromagnetic waves ranging in frequencies between **1 GHz – 300 GHz**.
- Mostly **uni-directional** i.e., sending and receiving antennas have to be aligned.
- **Cannot penetrate through walls.**
- Applications : unicast communications like cellular telephones, satellite networks, wireless LANs etc.



a. Dish antenna



b. Horn antenna

3. Infrared

- With frequencies 300GHz – 400THz.
- For short-range communication.
- High frequencies so cannot penetrate through walls.
- Prevents interference between one system and another.
- Applications : communications between devices such as keyboards, mouse, printers, PCs etc. ; short range communication in a closed area using LoS.



A student says: "Since single-mode fiber has only one mode, light travels faster than in multimode fiber." Which option is correct?

A. Correct

B. Incorrect. Single-mode fiber reduces dispersion, not necessarily the speed of light.

C. Correct because it has no reflections.

D. Correct because its refractive index is zero.



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A communication channel is perfectly noiseless. Increasing which parameter will NOT increase the maximum bit rate according to Nyquist?

- A. Bandwidth
- B. Signal Levels
- C. Noise
- D. Both A and B



A communication channel is perfectly noiseless. Increasing which parameter will NOT increase the maximum bit rate according to Nyquist?

- A. Bandwidth
- B. Signal Levels
- C. Noise**
- D. Both A and B



A student increases signal levels from 2 to 8 in a noisy channel. Will Shannon capacity necessarily increase?

- A. Yes
- B. No
- C. Only if bandwidth decreases.
- D. Only if attenuation increases.



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Why does multimode step-index fiber have greater modal dispersion?

- A. Light travels at different speeds.
- B. Different modes travel different distances at nearly the same speed.
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Why does graded-index fiber reduce modal dispersion?

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- B. Rays travelling longer paths move faster near the edge.
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A student says: "Single-mode fiber has no total internal reflection." Choose the correct statement.

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B. False. It still uses total internal reflection but supports only one propagation mode.

C. True because the light travels straight.

D. False because it has multiple modes.



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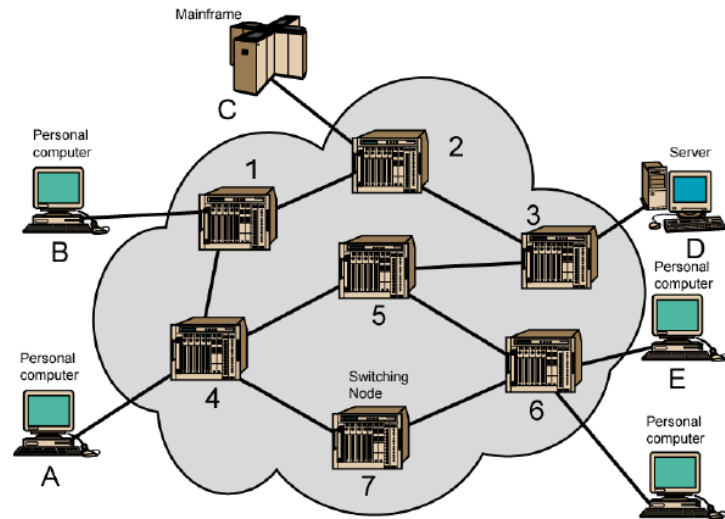
D. False because it has multiple modes.



SWITCHING NETWORKS

Switching Networks

- A collection of nodes and connections is a communications network
- Data is routed by being switched from node to node



How to Switch the Data?



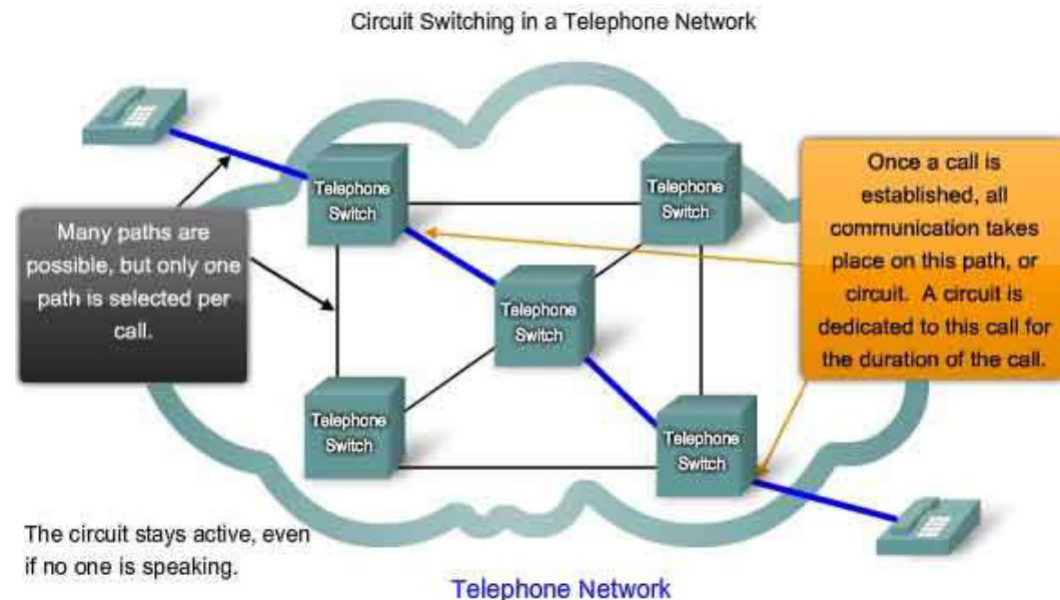
- Circuit switching
- Packet switching
 - ▣ Datagram
 - ▣ Virtual circuit

1. Circuit switching

- Communication via circuit switching involves three phases:
 - ▣ Circuit Establishment → connection oriented
 - ▣ Data Transfer
 - ▣ Circuit Disconnect
- Circuit → *Dedicated* communication path between two stations
- Developed for voice traffic (phone; Public Switched Telephone Network (PSTN))
 - ▣ Delay (and delay variation (jitter)) is guaranteed as required by voice traffic

Circuit switching

- ❑ Circuit switching is inefficient
 - ▣ Channel capacity dedicated for duration of connection → If no data, capacity wasted
- ❑ Information is transmitted at a fixed data rate with no delay (except for the propagation delay)



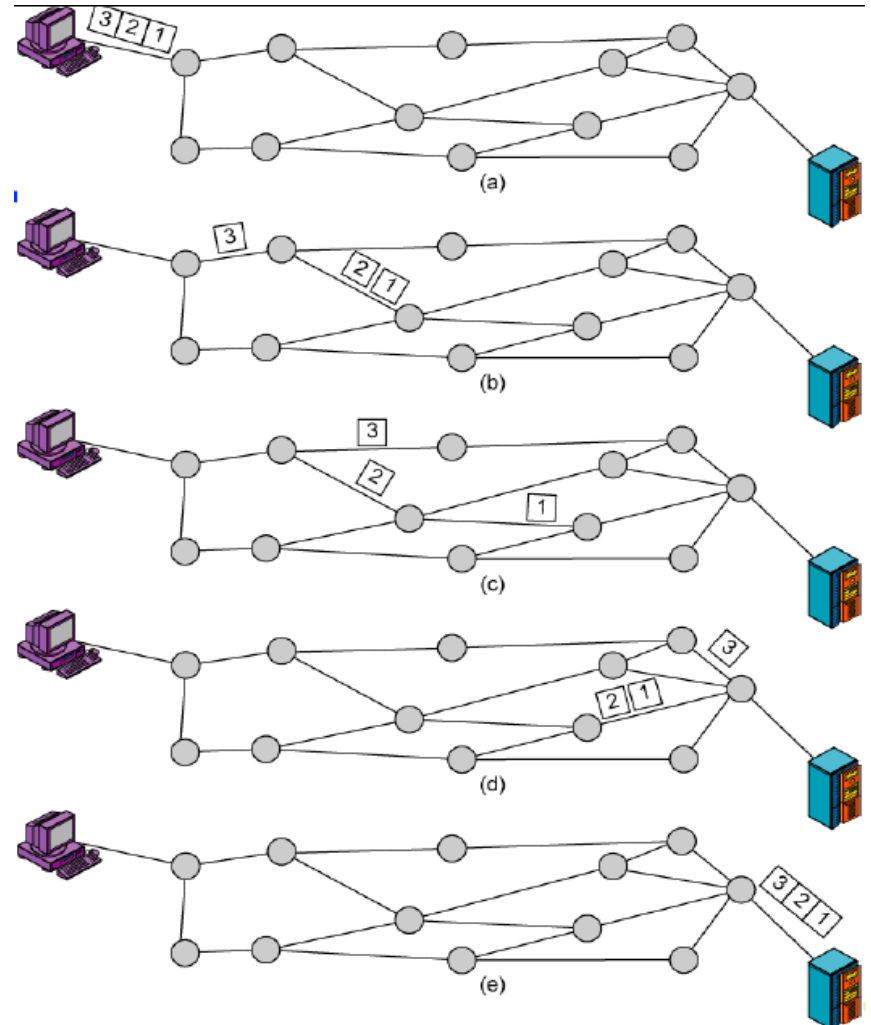
2. Packet Switching



- Station breaks long message into packets
- Packets sent one at a time to the network
- Packets handled in two ways
 - ▣ Datagram
 - ▣ Virtual circuit
- *Network resources are shared*

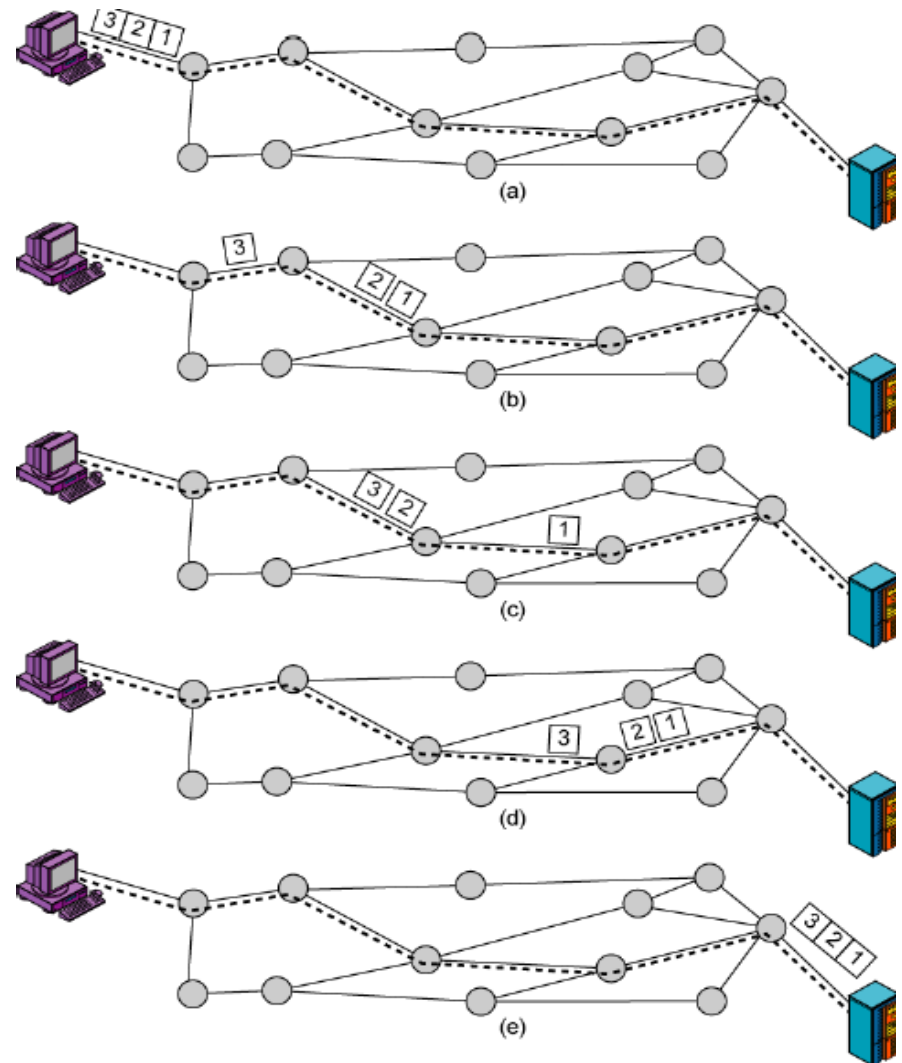
Datagram

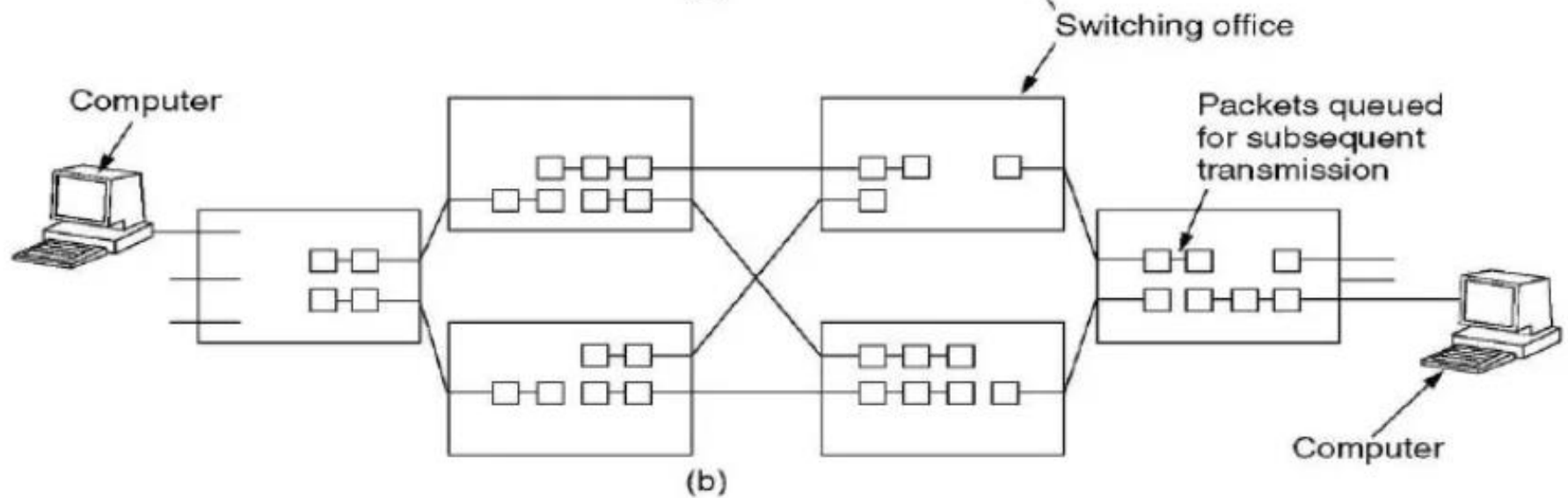
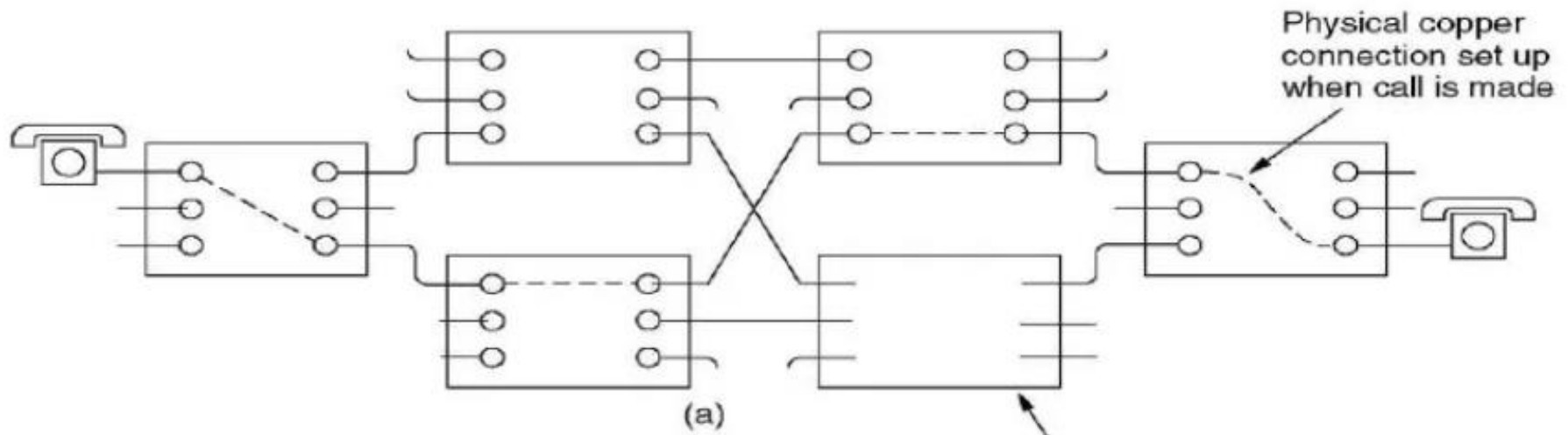
- *There's no preplanned route established* before any packets sent → connectionless
- Each packet treated independently
- Packets can take any practical route
- Packets may arrive out of order
- Packets may go missing
- Up to receiver to re-order packets and recover from missing packets



Virtual Circuit

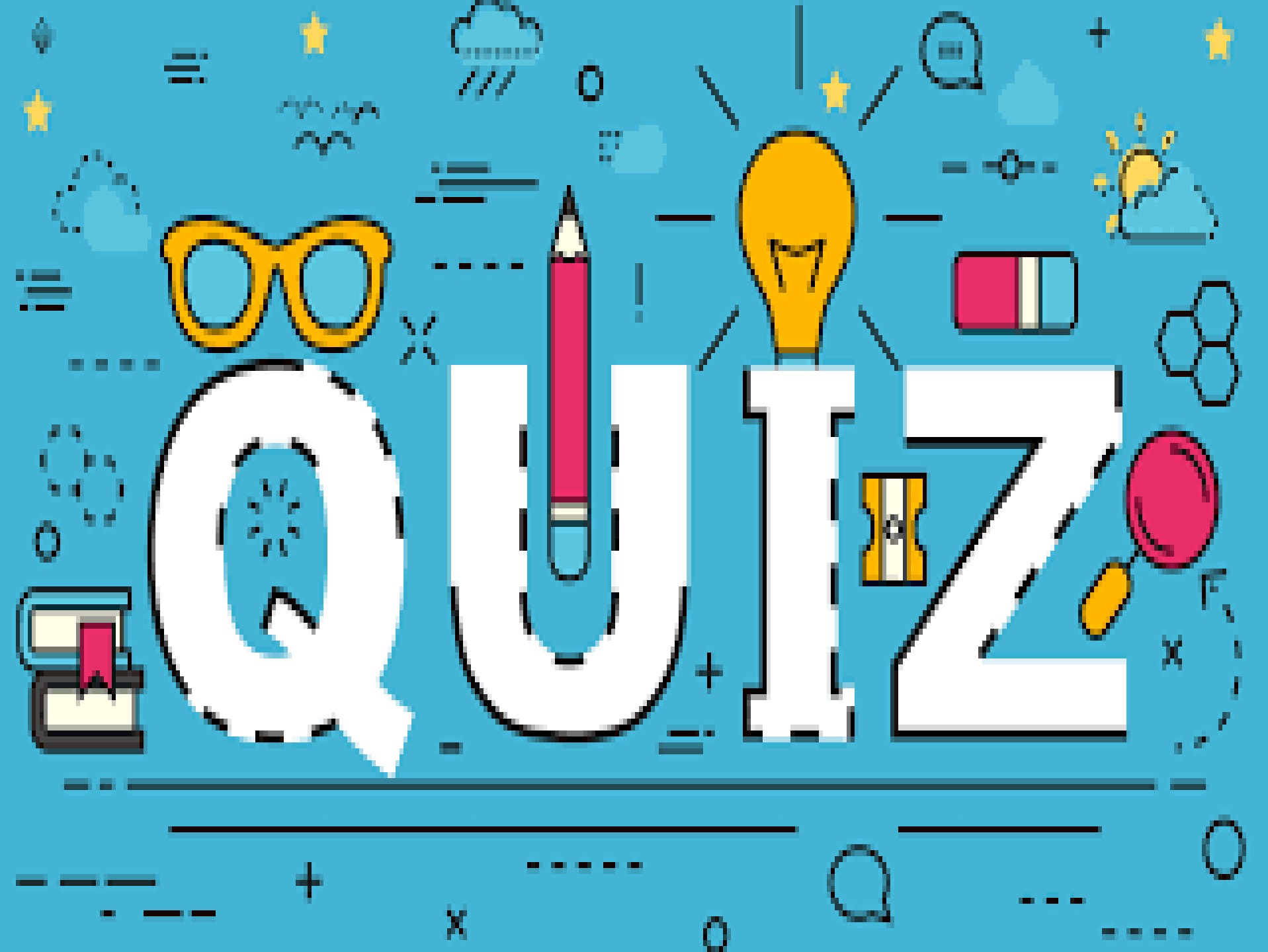
- *A preplanned route is established* before any packets are sent → connection oriented
 - ▣ Call request and call accept packets establish connection (handshake)
- Once route is established, all the packets follow same route
- Each packet contains a Virtual Circuit Identifier (VCI)
- No routing decisions required for each packet
- Clear request to drop circuit
- *Not a dedicated path*





(a) Circuit switching.

(b) Packet switching.





Which statement about Virtual Circuit Packet Switching is TRUE?

- A. It reserves bandwidth throughout communication.
- B. It reserves only a logical path.
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Which statement about Virtual Circuit Packet Switching is TRUE?

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Two users establish communication. During the session, no data is transmitted for one minute. Which switching technique still keeps the communication resources reserved?

- A. Datagram Packet Switching
- B. Circuit Switching
- C. Virtual Circuit Packet Switching
- D. Message Switching



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Why is packet switching generally more efficient than circuit switching?

- A. It uses analog signals.
- B. Resources are shared among users.
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- D. It reserves dedicated bandwidth.



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Thank You !